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COURSE MATERIAL

Sixty Years in Twenty Sections

CMSC 124 Activity 0

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Lesson 1 opens with vocabulary: readability, writability, reliability, cost, the handful of criteria we'll use all semester to argue about whether a language is any good. That vocabulary means nothing without cases to hang it on. Chapter 2 of Sebesta, "Evolution of the Major Programming Languages," is where the cases live: sixty years of language designers hitting the same walls and choosing different ways through them. Read it before lesson 1, so you have something to anchor the criteria on when we get to them.

The Reading

Sebesta, Chapter 2, sections 2.1 through 2.20. The whole chapter, start to finish. It runs from Zuse's Plankalkül in the 1940s to the markup-programming hybrids of the 2000s, by way of Fortran, Lisp, COBOL, SIMULA, Smalltalk, C++, Java, and a dozen others. Don't skip the genealogy chart (Figure 2.1). It's the one diagram that shows all twenty sections at once.

How to Read It

Don't read this chapter for dates. This isn't high-school history class. An assessment that asked "what year was COBOL released" would be checking whether you own a copy of the book. For every language section, look for three things instead:

1. What could existing languages not do? Fortran existed for a reason. COBOL existed for a different one.
2. What did this language add that didn't exist before? SIMULA 67 added classes and objects. Lisp added treating code and data as the same kind of thing.
3. Where did that feature go next? SIMULA's classes became object orientation as we know it in Smalltalk, C++, and Java. Almost nothing in this chapter is a dead end. Trace the line forward so the whole chapter turns into one continuous argument instead of twenty separate trivia entries.

Have fun reading! And please study very well for our first official lecture.